Miguel G. Silva

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Summary

PhD researcher and applied AI scientist. I take hard modeling problems (e.g., neural surrogates for physics simulation, multi-objective constrained optimization over engineered products) from formulation to production. Background in time-series modeling, pattern discovery, and interpretable ML, with first-author publications in Q1 journals.

Work Experience

2025 – Present | **Applied AI Scientist** at DareData Engineering

- Co-led the design of a production AI pipeline that automates custom cable design for a global automotive cable manufacturer, replacing a 16-week / ~€10k physical prototype-and-lab cycle with a 1-day / ~€1.8k software workflow (~80× faster, 5–10× cheaper per RFP).
- Formulated the surrogate-modeling layer: a Graph Neural Network with calibrated uncertainty, trained on multiphysics simulation history, turning hours of physics solves into millisecond inference.
- Designed the multi-objective constrained optimization over a mixed continuous/discrete design space (material × geometry × stack), balancing cost and performance under regulatory and manufacturability constraints. Calls the surrogate as a fast oracle, routing survivors to FEM validation.
- Designed the agent architecture for the LLM engineer assistant (OpenAI SDK): the specialist agents it can call (generation, simulation, optimization, proposal), the tools each exposes, and how the orchestrator routes intent.

2021 – 2026 | **Doctoral Researcher** at LASIGE & INESC-ID

- Analyzed smart meter data from 2,170 households using biclustering, uncovering statistically significant and actionable local water consumption patterns inaccessible to traditional clustering methods.
- Developed a scalable methodology to extract population density patterns from mobile phone data (3,743 cells) across Lisbon, uncovering multi-level seasonality and emerging urban trends.
- Mined motifs in multivariate time series; built `msig` to evaluate their statistical significance.
- Published 6 first-author papers in top-quartile (Q1) scientific journals.

2022 – 2024 | **Teaching Assistant** at IST, University of Lisbon

- Invited TA delivering 140+ hours on Information Retrieval (Python) and Database Systems (SQL) to 300+ students; 4.5/5 average rating across three semesters.

Education

2021 – 2026 **PhD in Computer Science**
Faculty of Sciences University of Lisbon, Portugal

2018 – 2020 **MSc in Data Science**
Faculty of Sciences, University of Lisbon, Portugal
GPA: 18 / 20 , Thesis: 19/20

Technical Skills

Languages & Core: Python · SQL · Git · LaTeX

ML / Deep Learning: PyTorch · Scikit-learn · Optuna · Pandas · NumPy

Applied AI: Surrogate Modeling · Multi-Objective Optimization · Time-Series Modeling · Uncertainty Quantification

LLM & Agents: OpenAI SDK · Tool Use · Structured Outputs · Orchestration

Engineering & MLOps: FastAPI · Pydantic · Docker · CI/CD (GitHub Actions) · MLflow · Async Pipelines

Publications

2026 **On Why and How Statistical Significance Criteria Can Guide Multivariate Time Series Motif Analysis** *Pattern Recognition Letters* | `PyPI: msig` | `GitHub`

2025 **Motif Mining in Multivariate Time Series** *Pattern Recognition* | `GitHub`

2025 **Next-Motif: Predicting Irregular Pattern Recurrence in Time Series** *Springer (book chapter)* | `GitHub`

2024 **A Comprehensive Survey on Biclustering-based Collaborative Filtering** *ACM Computing Surveys*

2023 **Actionable descriptors of spatiotemporal urban dynamics from large-scale mobile data** *Environment and Planning B* | `GitHub`

2022 **Water consumption pattern analysis using biclustering: When, why and how** *MDPI Water*